

Fundamentals of Data Center Networking

What AI Workloads Need from the Network

A technical overview for network engineers and architects building AI infrastructure



The Big Picture

Modern AI workloads fundamentally transform traditional network design requirements:

East-West Dominance

AI workloads generate massive node-to-node traffic flows, unlike traditional north-south patterns

Critical Performance Metrics

Throughput, low latency/jitter, and losslessness become non-negotiable requirements

Hidden Bottlenecks

Bisection bandwidth measurement reveals true fabric capacity limitations

Scale-Out Architecture

Design must accommodate horizontal growth while maintaining predictable performance



Clos/Spine-Leaf Underlay



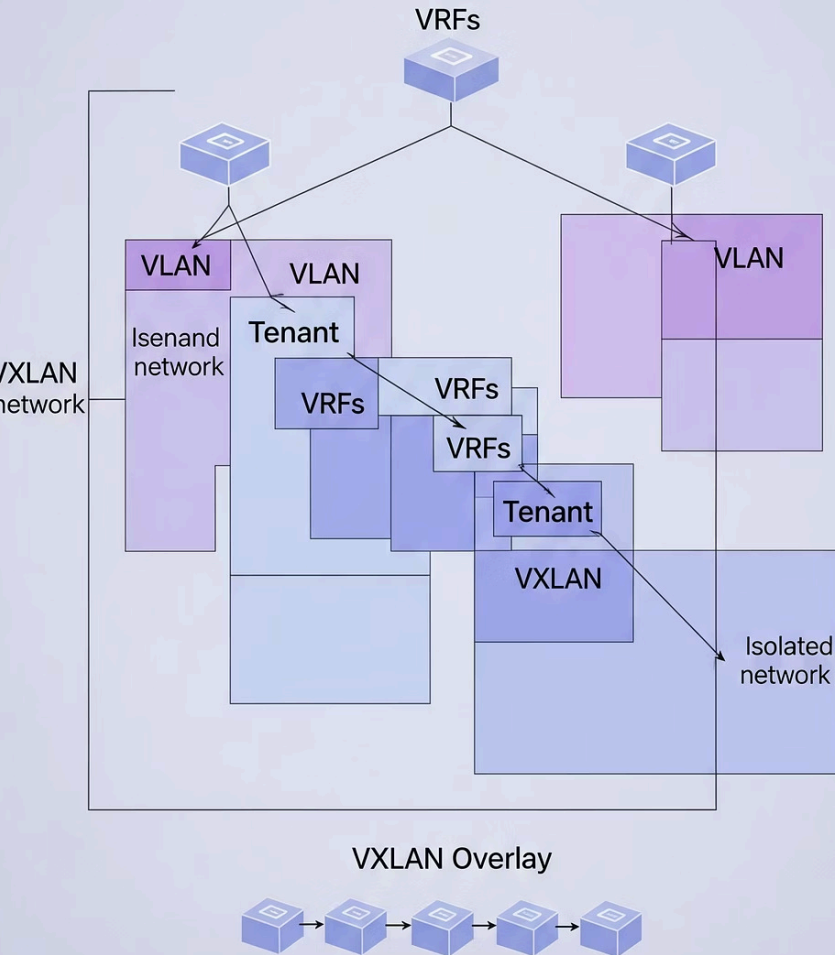
Core Architecture

- **Leaf switches** connect directly to servers
- **Spine switches** interconnect all leaf switches
- Layer-3 fabric with **ECMP (Equal-Cost Multi-Path)** routing
- Typically 2-3 hops maximum between any two endpoints

Implementation Considerations

- Plan **oversubscription ratio** based on workload (1:1 for AI training)
- Achieve modular scale without redesigning the fabric
- Isolate faults to minimize blast radius
- Ensure deterministic performance with consistent hop count

Network Segmentation



L2/L3 Segmentation & Overlays

Layer 2: VLANs

Basic segmentation for smaller deployments

- Limited scale (4,094 VLANs maximum)
- Broadcast domain isolation
- Simplest implementation

Layer 3: VRFs

Tenant isolation at the routing layer

- Separate routing tables per tenant
- Improved scale compared to VLANs
- Maintains routing separation

VXLAN + EVPN

Scalable overlay for large multi-tenant DCs

- L2 over L3 encapsulation
- MP-BGP EVPN control plane
- Thousands of virtual networks

Micro-segmentation

Fine-grained per-application security

- Workload-level policies
- Zero-trust implementation
- East-west traffic control

Transport & Congestion Control

Transport Protocols

Choose the right protocol for your workload:

- **TCP:** General traffic, reliable delivery
- **RDMA/RoCEv2:** Ultra-low latency, high throughput, critical for AI workloads

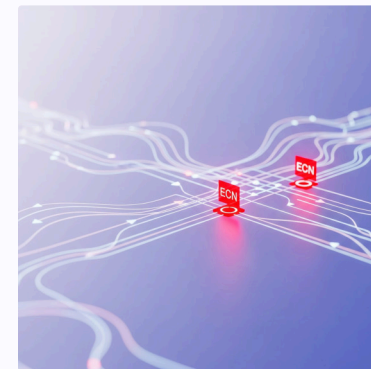
Congestion Management

- **PFC (Priority Flow Control):** Creates lossless traffic classes
- **ECN/RED:** Early congestion notification
- **DCQCN:** Data Center Quantized Congestion Notification

Optimization Techniques

- **Jumbo frames:** Reduce CPU/interrupt overhead (9000 MTU)
- **Buffer tuning:** Balance between latency and packet loss
- **End-to-end QoS:** Consistent marking and treatment

Warning: PFC without proper ECN/DCQCN can lead to "pause propagation storms" that collapse network performance. Always implement congestion control alongside PFC.



Links & Interconnect Choices

Ethernet Technologies

- 25/50/100/200/400/800 GbE
- QSFP/OSFP form factors
- Broad ecosystem support
- Standard for most deployments

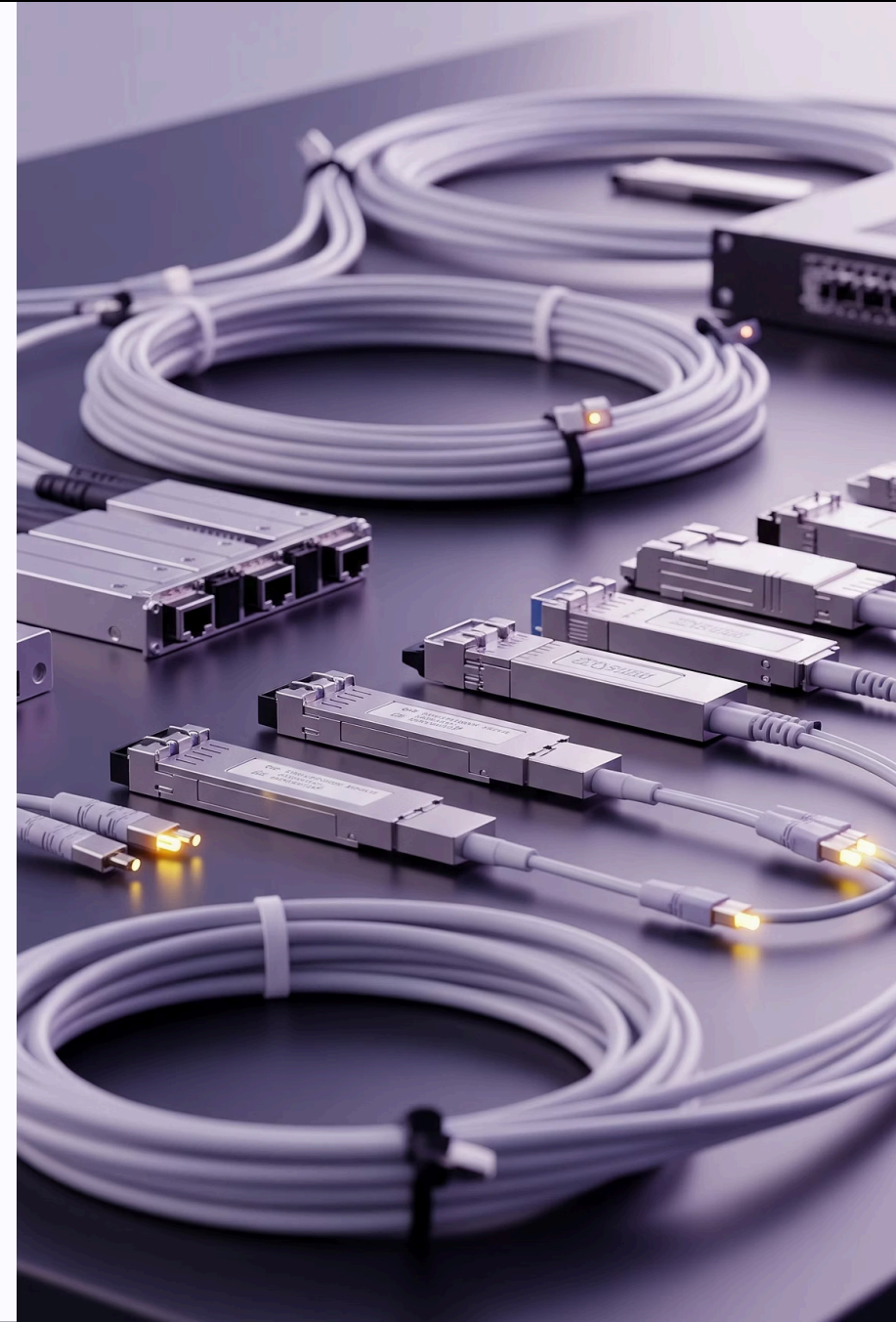
InfiniBand

- HDR (200 Gbps) / NDR (400 Gbps)
- Ultra-low latency (300-600ns)
- RDMA native, optimized collectives
- Preferred for HPC/AI clusters

Cabling Considerations

- **DAC:** $\leq 5\text{m}$, lowest cost/power
- **AOC:** $\leq 30\text{m}$, moderate cost
- **Optics:** Up to 10km+, highest cost
- Standardize for operational simplicity

The choice between Ethernet and InfiniBand often depends on application requirements, existing infrastructure, and team expertise. Many AI clusters use InfiniBand for compute and Ethernet for storage/management.



AI Workload Patterns

Dominant Communication Pattern: All-Reduce

Parameter synchronization creates bursty, many-to-many traffic flows that stress network bisection bandwidth



Framework Implementations

- **NCCL:** NVIDIA Collective Communications Library
- **Gloo:** Facebook's open-source collective comms
- **MPI:** Message Passing Interface (traditional HPC)
- **Horovod:** Uber's distributed training framework

Topology Considerations

Different collective operations perform differently based on network topology:

- Ring algorithms scale linearly (N)
- Tree algorithms scale logarithmically ($\log N$)
- Mesh/all-to-all operations stress bisection

Co-design Factors

- Node placement strategy
- NUMA boundaries and awareness
- NIC/GPU affinity for direct data transfer
- Job scheduling aligned with topology

Service Networking & Load Balancing

Load Balancing Approaches

L4 (Transport): DSR/Anycast for massive scale

L7 (Application): Content-based routing, authentication, TLS termination

In-cluster Networking

CNI: Container Network Interface plugins

eBPF: Extended Berkeley Packet Filter for high-performance datapaths

Service Mesh: Optional mTLS, traffic management, observability

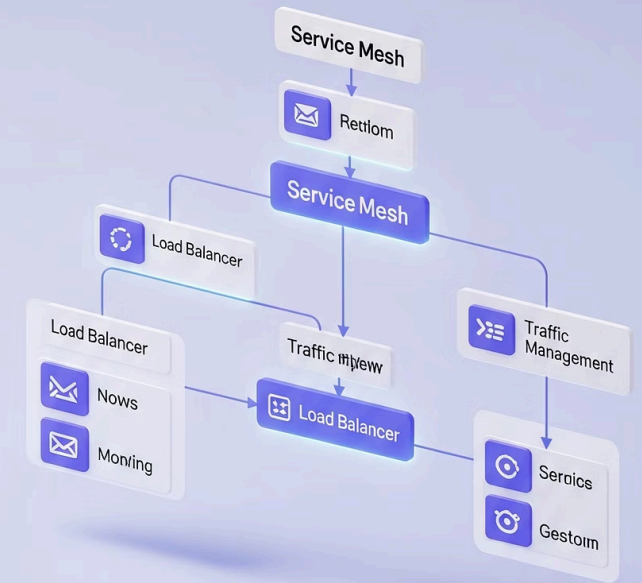
Global Traffic Management

Anycast: Same IP advertised from multiple locations

Ingress/Egress: Smart traffic routing based on application semantics

Protection: Rate limiters, circuit breakers for backend services

Modern AI deployments often run as microservices alongside traditional applications, requiring sophisticated service networking. Well-designed service networking improves reliability, security, and operational simplicity.



Observability & Troubleshooting

Data Collection Methods

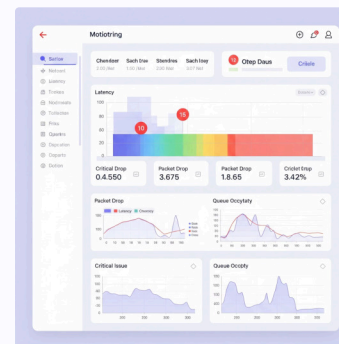
- **Streaming telemetry:** Real-time, subscription-based
- **sFlow/NetFlow/IPFIX:** Traffic sampling and analysis
- **SNMP:** Traditional polling for baselines

Key Performance Indicators

- **Latency heatmaps:** Visualize outliers and patterns
- **Packet drops:** Zero tolerance for AI workloads
- **Queue occupancy:** Early indicator of congestion
- **Buffer utilization:** Microbursts and congestion

Troubleshooting Toolkit

- **ping/tracepath:** Basic connectivity
- **iperf3:** Throughput validation
- **SPAN/ERSPAN:** Port mirroring for deep analysis
- **tcpdump/Wireshark:** Packet-level inspection



Correlate application metrics (GPU utilization, training throughput) with network KPIs to identify performance bottlenecks.



Security & Resilience

Zero-Trust Security



- Access Control Lists (ACLs) at multiple layers
- Micro-segmentation between workloads
- East-west firewalls for lateral movement prevention
- Authentication for all network connections

Link & Path Resilience



- LACP/MLAG for link aggregation
- Fast reroute capabilities within the fabric
- Equal-Cost Multi-Path (ECMP) for load distribution
- Automated failure detection and recovery

Advanced Protection



- DDoS mitigation at network edge
- Egress filtering and traffic policies
- MACsec/IPsec for encryption in transit
- Regular chaos engineering drills

High-value AI models and data require robust security controls without compromising performance. Integrate security into the initial design rather than retrofitting it later.

Design Best Practices



Keep the Underlay Simple

Layer-3 fabric with ECMP provides the foundation; push complexity to overlay services

Align End-to-End Parameters

MTU, buffer configurations, and QoS must be consistent across all network elements

Separate Traffic Classes

Implement distinct policies for control, storage, AI training, and inference workloads

Document Everything

Maintain detailed IPAM, routing policies, and failure playbooks for operational excellence

❏ **Remember:** Network design for AI is iterative. Start with a solid foundation, measure actual workload performance, and refine as needed.



Key Takeaways



Modern DC Architecture

Spine-leaf topology with L3/ECMP is the proven foundation for AI infrastructure



Congestion Management

PFC/ECN/DCQCN is critical for handling bursty all-reduce operations in AI training



Measurement is Key

Optimize based on bisection bandwidth and jitter metrics - not just raw throughput



Security & Resilience

Build secure, resilient, and measurable networks to power reliable AI operations

Questions? Contact our network architecture team at networking@company.com